



COURSE SYLLABUS						
Course Code	INTE 403	Course Title	Systems Administration and Maintenance		Course Credit	3
Course Description		The course Systems Administration and Maintenance provides students with the knowledge and hands-on skills required to manage, configure, and maintain modern IT infrastructures. It covers operating systems installation, network services configuration, user and security management, virtualization, containerization, cloud computing, and automation techniques. Emphasis is placed on applying best practices to ensure system reliability, security, and scalability in enterprise environments. Through practical labs and projects, students will gain the ability to design and administer integrated systems using current technologies and tools. This course prepares learners for real-world roles in IT operations, system administration, and infrastructure management.				
Pre-Requisites		INTE 301	Co-Requisites	None		
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				



INSTITUTIONAL
LEARNING OUTCOMES
(ILO)

- As a polytechnic state university, PUP shall develop its students to possess:
1. **Critical and Creative Thinking.** Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world.
 2. **Effective Communication.** Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances.
 3. **Strong Service Orientation.** Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors.
 4. **Adept and Responsible Use or Development of Technology.** Graduates demonstrate optimized and responsible use of state- of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills.
 5. **Passion for Lifelong Learning.** Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.



	6. Leadership and Organizational Skills. Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills. 7. Personal and Professional Ethics. Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances. 8. Resilience and Agility. Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment. 9. National and Global Responsiveness. Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.
COLLEGE GOALS AND OBJECTIVES	The Polytechnic University of the Philippines – Taguig Campus aims to achieve the following goals: 1. Offer curricula which are relevant and responsive to the changing needs of the industry and society; 2. Provide trainings to students that will enable them to become potent instruments to socio-economic development. 3. Prepare students to become productive, creative, innovative, and dynamic in their chosen fields of specialization. 4. Provide state-of-the-arts facilities of learning to optimize student development; 5. Inculcate appropriate values that are necessary to build a humane, disciplined, nationalist, and independent society; 6. Tap potentials of students, faculty, administrative staff and other stakeholders in formulating policies for institutional development; 7. Involve all stakeholders in enhancing programs for student’s faculty and employees’ development; 8. Partner with industry and other research institutions strengthening research capabilities of faculty, employees and students; 9. Facilitate presentation of research outputs in international fora, and their publication in recognized journals; 10.Undertake outreach and research-based extension programs by tapping the expertise and other resources of all stakeholders and, 11. Produce technologies for commercialization of livelihood improvement.

PROGRAM LEARNING	Based on CMO	Alignment to ILOs								
		1	2	3	4	5	6	7	8	9
	Apply knowledge of computing, science, and mathematics appropriate to the discipline.									



OUTCOMES (PLO)	Analyze complex problems, and identify and define the computing requirements appropriate to its solution												
	Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems.												
	Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.												



	Integrate IT-based solutions into the user environment effectively.												
	Assist in the creation of an effective IT project plan.												
	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession												
	Understand best practices and standards and their applications.												
	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.												
	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.												
	Analyze the local and global impact of computing information technology on individuals, organizations, and society.												
	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.												
	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development												

COURSE LEARNING OUTCOMES (CLO)	Explain the fundamental concepts, roles, and responsibilities of systems administration within modern IT infrastructures.	Alignment to PLOs							
		1	2	3	4	5	6	7	8
	Install, configure, and manage operating systems and network services to ensure reliable and secure IT environments.								
	Demonstrate user, group, permission, and security management to protect system resources and data integrity.								
	Apply virtualization technologies to optimize computing resources and support scalable infrastructure.								



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	Deploy and manage containerized applications and orchestration platforms (e.g., Docker, Kubernetes) for efficient system operations.								
	Implement serverless and cloud computing solutions (IaaS, PaaS, SaaS) for flexible and cost-effective system administration.								
	Automate administrative tasks using scripting and configuration management tools to improve system performance, backup, and recovery operations.								
	Design and present an integrated enterprise-grade system infrastructure project demonstrating the application of systems administration principles, cloud technologies, containerization, and automation.								

General Santos Avenue, Lower Bicutan, Taguig City, Philippines 1632
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Website: www.pup.edu.ph | Inquiries: https://bit.ly/PUPSINTA

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WORLD
UNIVERSITY
RANKINGS

2025

7th in the Philippines
92nd in Southeast Asia
541-560 in Asia



WORLD
UNIVERSITY
RANKINGS
for INNOVATION

Rank 21
Rank 39
Rank 50



STARS

5★

5★

5★

5★

5★



Times Higher Education
Impact Rankings 2024

5★

Employability

Teaching

Online Learning

Academic Development

Internationalization

Social Responsibility

Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)

Inclusiveness



OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/Topics	Instructional Delivery Design		Assessment		Alignment to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)	Face-to-Face	FLTAs	1	2	3	4
1	- Equip students on the pivotal ideas embedded in the vision, mission, and 10 Pillars of PUP - Gain an overview of the course subject and requirements	- Orientation about PUP Vision, Mission, and 10 Pillars - Overview about the Subject Course and Course Requirements	Present the VMGO, policies, and guidelines on the conduct of Face-to-Face; and of the requirements of the course.		Formative Assessment (Sharing of Learning)		X			
2	- Define systems administration and its role in IT infrastructure. - Explain system architecture and components.	- Overview of System Administration, Roles & Responsibilities, System Lifecycle - Hardware and Software Architecture, OS fundamentals, Command-line basics	- Lecture, discussion on sysadmin duties - Hands-on: Exploring OS architecture	- Virtual orientation, course overview - Virtual lab walkthrough	- Diagnostic quiz - Short quiz, lab report	- Create an infographic summarizing SysAdmin responsibilities - Submit a short reflection (250–300 words) discussing how system components interact in real-world setups				



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3	Install and configure operating systems.	Installation processes (Windows/Linux), Partitioning, File systems	Demo: Dual boot / VM install	OS install via virtual lab	Step-by-step video guide	Record and upload a short video (max 3 min) demonstrating your OS installation process using a VM				
4	Manage users, groups, and permissions.	User management, Access control, File permissions	Lab: User/group management in Linux	Command-line management session	Practice exercises	Create a short tutorial or infographic showing Linux permission symbols and examples				
5	Configure networking and troubleshoot connectivity issues.	IP addressing, DNS, DHCP, routing, troubleshooting	Lab: Network setup using VMs	Network simulation via Packet Tracer	Networking practical exam	Submit a network topology diagram using online tools (e.g., draw.io or Lucidchart) with labeled components				
6	Perform system monitoring and maintenance.	Monitoring tools (top, htop, Task Manager), Logs, Backups	Lab: Log analysis, backup scripts	Discussion: Automating maintenance tasks	Maintenance script submission	Write a short blog post (200–300 words) explaining how automation improves maintenance efficiency				
7	Understand and implement system security measures.	Firewalls, Security policies, SSH, Encryption	Firewall configuration exercise	Discussion: Security best practices	Security audit checklist	Create a cybersecurity awareness poster (digital format) emphasizing system security principles				



8	- Understand and implement system security measures. - Apply virtualization concepts for resource optimization.	- Firewalls, Security policies, SSH, Encryption - Virtualization overview, Hypervisors (Type 1/2), VirtualBox, VMware	- Firewall configuration exercise - Lab: Creating and managing VMs	- Demo: Hypervisor management - Walkthrough: Docker Hub, Dockerfile	- VM configuration project - Docker lab assessment	- Write a short comparative report on VMware vs. VirtualBox (advantages, limitations, use cases) - Create and share a Dockerfile with documentation on how to build and run your custom container				
9	MIDTERM WRITTEN EXAMINATION									
10	Manage container orchestration systems.	Kubernetes fundamentals, Pods, Services, Deployments	Lab: Setting up a local K8s cluster (Minikube)	K8s architecture discussion	Quiz & mini project (K8s deployment)	Develop a YAML deployment file and submit a short explanation video (1–2 min) describing its purpose				
11	Understand microservices architecture.	Microservices overview, APIs, Service communication	Lecture with real-world examples	Demo: REST API-based microservice	Reflection paper	Submit a concept map comparing monolithic and microservice architectures				
12	Implement serverless computing concepts.	Serverless model, AWS Lambda / GCP Functions / Azure Functions	Demo: Simple serverless function	Online simulation of serverless platform	Serverless exercise submission	Create a short write-up on a real-world company successfully using serverless computing (with sources cited)				
13	Apply cloud computing in systems administration	Cloud models (IaaS, PaaS, SaaS), Deployment, Cost models	Cloud setup using free-tier account	Live demo: Deploy app on cloud	Cloud deployment project	Submit screenshots and a short report (max 1 page) describing your cloud				



	.					instance and deployment process				
14	Automate system administration tasks.	Scripting (Bash, PowerShell), Configuration management (Ansible)	Lab: Create automation scripts	Demo: Infrastructure as Code (IaC)	Script automation submission	Write a brief documentation (300 words) on your script explaining its logic and automation purpose				
15	Implement system backup and recovery strategies.	Backup methods, Disaster recovery, Redundancy	Demo: Backup & recovery scenario	Discussion: Business continuity planning	Backup/recovery lab	Create a disaster recovery flowchart showing backup strategies and recovery timelines				
16	Monitor system performance and optimize operations.	Resource monitoring, Performance tuning, Alerts	Lab: Monitor CPU, memory, disk usage	Walkthrough: Monitoring dashboards (Prometheus/Grafana)	Monitoring dashboard report	Submit monitoring screenshots and write a short interpretation of the collected metrics				
17	- Integrate all learned concepts in a real-world setup. - Evaluate and present system administration project outcomes.	- Capstone project integration – hybrid system with cloud + containers - Project presentation, Peer evaluation, Reflection	- On-site support and project consultation - Final presentation and defense	- Online project mentoring - Group reflection and feedback	- Capstone project submission - Final project evaluation & reflection paper	- Submit project documentation draft and peer evaluation using an online feedback form - Submit a recorded presentation or portfolio including screenshots, diagrams, and lessons learned				
18	FINAL EXAMINATION ()									



		Resources and Applying Strategies in Teaching Prose <i>Competency 4:</i> Recognizing Assessment in Teaching Prose	achievement which are consistent with the selected competencies • Learning Plan Making according to the English curricula that develops higher order thinking skills of learners through the use of prose and dramatic texts		• Quizzes					
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FINAL EXAMINATION

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CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to-face sessions.	Students' Responsibilities Students are expected to: a. Take time to read and watch the provided class learning materials. b. Fulfill class requirements such as posting on discussion boards and submitting group presentations. c. Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.



The instructor may not be able to remind all students about their missing/unsubmitted requirements. If at the end of the course students still have missing requirements, the instructor will compute for the final grade without those submissions (They will be graded with 0) and submit the grade. If the final grade is “INC” grade, the student must contact the instructor to decide in completing the course, ASAP.

d. Confer with classmates or the instructor about the discussions during the synchronous sessions that they missed if unable to attend the live sessions. Recordings will be provided but may not be available immediately.

e. Inform the instructor of concerns that may prevent them from fully participating in the class. This includes withdrawal from the course, long term illnesses, personal matters, and others. Teachers will not be able to help you if you don’t tell them of these concerns.

Attendance.

We recognize that students have varying access levels to technology and connectivity, which is why for the most part, the course is conducted asynchronously. During synchronous sessions, there will be no checking of attendance. However, everyone is encouraged to participate in these sessions. A recording and/or transcript of synchronous classes will be made available for students who are unable to attend a synchronous session.

Late Submissions.

Deadlines will be set for the submission of course requirements. This allows the instructor sufficient time to review them. It also helps students pace their work. If you experience problems submitting your work on time, please contact the instructor ahead of time. All requirements must be completed to pass the course. Late submissions will not merit deductions but may not receive timely feedback.

Cheating and Plagiarism.

Any work submitted by a student in this course for academic credit will be the student’s own work. Any form of academic dishonesty will be fully dealt with according to the stipulations of the PUP Student Handbook.

Participation.

Participation during discussions is greatly encouraged, and students should feel free to ask questions when an idea or lesson is unclear or confusing, especially during synchronous sessions.

Announcements.

Class announcements will be communicated through the class president. Other communication shall always be coursed through university-issued (i.e., official) email accounts. Students must use their PUP student accounts.

Intellectual Property and Copyright.

Original materials created for this online class is covered by university policies on intellectual property. Some (third-party) instructional materials (e.g., YouTube videos) might be under the public domain or are open access, while others (e.g., some assigned readings) might be subject to fair use. All third-party instructional materials will be attributed to their original



creators. Electronic copies of the assigned readings are for your private consumption only. Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be available for only a



Mid Term	Final Term
Class Standing 70% Exercises/Activities Summative Quizzes Recitation and Attendance Reflection Paper Midterm Examinations 30% 100 %	Class Standing 70% Exercises/Activities Summative Quizzes Recitation and Attendance Reflection Paper Final Examinations () 30% 100%

	limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.
COURSE REQUIREMENT/S	GRADING SYSTEM



- Students must actively participate in class discussions and laboratory sessions, ensuring regular attendance and punctuality. They are expected to complete quizzes and written assessments to demonstrate their understanding of key concepts.
- Students must also accomplish a capstone or final practical assessment where they will design, implement, secure, and maintain a complete system environment, showcasing their overall competency in systems administration and maintenance.

Final Grade = (Midterm Grade + Final Term Grade) / 2

CLASS INFORMATION		FACULTY INFORMATION	
Course, Year and Section: BSIT 4-1		Name of Faculty: MODESTO, LADY MELINDA MINETTE LOPEZ	
Day and Time: Saturday 1:00 PM - 6:00 PM		Consultation Time:	
Room: A-304		Office Tel. No./ Mobile Phone No.	
School Year: 2526	Semester: 1st Semester	E-Mail Address:	



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Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
LADY MELINDA MINETTE L. MODESTO Faculty Member	MS. DIANNE S. SEGURORA Head of Academic Programs	 DR. RHYAN V. MOLINAR Head of Academic Programs	DR. ARMANDO A. TORRES Vice President for Campuses

General Santos Avenue, Lower Bicutan, Taguig City, Philippines 1632
Direct Line: (02) 8837 5858 to 60 | Email: taguig@pup.edu.ph
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